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A Spurious-Free Source-Model Technique for Waveguide Mode Determination

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Abstract

Some difficulties in the application of source-model techniques to waveguide mode determination are pointed out, and shown to stem from approximate spurious solutions to the characteristic equation of the waveguide. An approach that avoids these spurious solutions is presented, yielding a method more reliable and efficient.

1 Introduction

The Source-Model Technique (SMT) can be classified as a Method of Moments (MoM), where the fields of discrete sources, located a distance away from the boundaries, are used to approximate the fields (see [1] for a review). This is in contradistinction to the probably more common, surface-formulations, that employ continuous sources coincident with the boundaries. The SMT has been applied to a wide range of scattering problems and is known in this context also by many other names: "Method of Auxiliary Sources" [1], "Generalized Multipole Technique" [2], "Method of Fundamental Solutions" [3], and so on. Simplicity of implementation, high numerical accuracy and stability, make the SMT a viable alternative to the surface-formulations.

Since waveguide mode determination can be thought of as a scattering problem with vanishing excitation, it is natural to assume that the SMT can be applied to it straightforwardly. The literature on this subject, however, is quite scarce. Indeed, only recently has the SMT been used to calculate the modes of an elliptical metallic waveguide [3].

Although the SMT is a powerful method for waveguide mode determination, we will show that the existence of spurious solutions can hinder its practical application considerably.

This letter is organized as follows: Section 2 outlines the essentials of the SMT and describes how the existence of spurious solutions in the SMT formulation renders the determination of modal solutions problematic. Section 3 explains the problematic behavior by use of an analytic approximation of the spurious solutions. Section 4 presents the proposed remedy, together with a